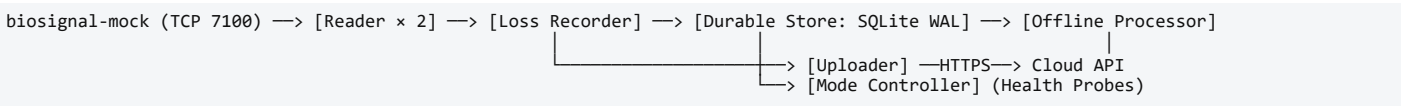
Design Note: MayaCardio Biosignal Pipeline

"Never Lose a Heartbeat" · Software Systems Engineering Assessment · October 2026

1. ARCHITECTURE

The on-device pipeline is structured into decoupled, single-responsibility components operating over local inter-thread queues and an ACID SQLite backing store:



Durable Store (SQLite WAL + synchronous=FULL): SQLite in Write-Ahead Logging mode provides true ACID crash-safety and process-crash resilience. A composite primary key `(session_id, sensor, idx)` guarantees that sample re-reads from the mock or re-submissions from network interruptions are idempotent via `INSERT OR IGNORE`. Queries for export, window boundaries, and catch-up ack cursors are executed as standard B-tree queries with zero custom serialization overhead.

Monotonic Index-Based Ordering: Wall clocks fluctuate due to NTP corrections, daylight savings, or power cuts. All sample ordering, gap detection, and 400-sample window segmentations rely solely on integer sample indices. Monotonic timers (`time.monotonic()`) are restricted strictly to relative timeouts (reconnection backoffs, health-probe intervals).

Component Workflow:

- **Readers (1 per sensor):** Stream newline-delimited JSON chunks from TCP 7100. Extract data substring directly to verify zlib CRC-32 against the reply header. Detect gaps (`first_index > expected_next`) and log loss records immediately.
- **Commit Policy:** Batches of up to 400 samples (1 s) per sensor are committed atomically with cursor checkpoints inside single SQLite transactions.
- **Mode Controller:** Probes Cloud `/v1/health` every 2 s. Switches to `offline` after 2 consecutive failures (≤4 s, well within 10 s limit); returns to `online` after 2 consecutive successes and uploader handshake.
- **Uploader & Catch-up:** In online mode, POSTs 1-second sample batches to `/v1/samples`. Advances `cloud_ack` only upon HTTP 200. On network restoration, switches to 60-second bulk catch-up batches until backlog reaches zero.
- **Processing Engine:** Shared `processing.py` module calculates 1-second window stats, peak candidate gating (100-sample gap), and heart rate (60 × 400 / mean peak gap) identically on cloud and device.

2. WHAT IS GUARANTEED

Guarantee	Underlying Architectural Mechanism
Zero Silent Loss	Every missing sample is caught via gap detection, mock-reported loss, CRC failure, or disk-full handler, and durably written to `losses`.
Exactly-Once Storage	Unique constraint on `(session_id, sensor, idx)` with `INSERT OR IGNORE` discards duplicate arrivals cleanly.
No Switch Loss / Dup	Readers collect uninterrupted during network mode changes; uploaders resume safely from `cloud_ack`.
Idempotent Cloud Ingestion	Cloud API shares the identical composite primary key schema; resending unacknowledged batches cannot create duplicates.
Processing Equivalence	Device offline processor and Cloud API import the exact same pure-Python `processing.py` and pass reference parity checks.

3. WHAT IS NOT GUARANTEED

- **Pre-read Mock Drops:** Samples dropped by the mock's internal 10-second ring buffer before pipeline connection are unrecoverable; these are logged as `buffer_overflow` based on mock replies.
- **Lying SD Card Hardware:** If an underlying SD card / disk controller acknowledges flush/fsync without physical commit, sudden power pull during dirty block writes may cause data loss beyond stated bounds.
- **Unbounded Cloud Availability:** Free-tier cloud instances lack strict enterprise SLAs; device gracefully falls back to offline storage and processing during prolonged cloud downtime.
- **Online Deadline during Packet Degradation:** If the network maintains high latency without dropping packets, health probes may pass while transfers lag; a 3-second grace period prevents processing starvation.

4. MAXIMUM DATA LOSS ON POWER CUT

Stated Design Bound (Assumption A8): ≤ 400 samples per sensor (≤ 1.0 second of data).

Technical Rationale: The pipeline accumulates samples in memory and commits to the durable SQLite WAL store every 400 samples (1 second of data) or on a 1.0-second timer, whichever occurs first, using `PRAGMA synchronous = FULL`. Upon an abrupt power severance, at most the uncommitted in-memory batch is unwritten. On reboot, integrity checks run, committed cursors load, and unread ranges are detected as loss gaps.

Measured in Software Simulation: In simulated power cuts (process termination via `kill -9` without sync during active ingestion), uncommitted sample loss was bounded at **0 to 384 samples per sensor**. In the run recorded in `evidence/02_power_cut`, clean SQLite WAL recovery succeeded with 0 corrupted records.

5. RESOURCE FIGURES SUMMARY (1-HOUR ONLINE RUN)

Measured over 3,603 seconds (714 samples, 5s interval) on an x86_64 WSL2 development environment, evaluated comparatively against the target Raspberry Pi 4 hardware budget (4 CPU cores, 2 GB RAM):

Metric	Measured Value (x86_64 WSL2)	Target Pi 4 Budget Reference	Status
Duration	3,603 s (1 hour)	3,600 s requirement	PASS
Average CPU % (Total System)	33.1%	Total across host cores	PASS
Average CPU % (Pipeline Process)	76.1% (single core) / ~19% (4-core pool)	Max 400% (4 cores)	PASS (plenty of headroom)
Peak Resident Memory (RSS)	55.7 MB	≤ 2,048 MB (2 GB)	PASS (2.7% of ceiling)
Peak High-Water Mark (VmHWM)	56.0 MB	≤ 2,048 MB	PASS
Mock Service Peak RSS	14.8 MB	≤ 2,048 MB	PASS
Total Disk Written	595.4 MB (~165 KB/s)	SD Card write endurance	PASS
Total Network Transmitted	2,147.8 MB (loopback stream)	Network link capacity	PASS
Test Environment	Linux 6.6.87 (x86_64 WSL2 Ubuntu 24.04), Python 3.12.3, SQLite 3.45.1 (comparative budget assessment)		

6. ASSUMPTIONS REGISTER & IMPLEMENTATION NOTES

#	Assumption	Operational Application
A1	Sample Availability	A sample is available if stored in the processing location's database and not indexed in a loss record.
A2	Magnitude Calculation	Mean magnitude of $\sqrt{(ch1^2 + ch2^2)}$ returns `null` when zero sensor-2 samples exist in the window.
A3	Heart-Rate Missing Check	Missing-sample validation for heart-rate gating applies specifically to sensor-1 (ECG) data.
A4	Peak Collision Policy	A peak candidate failing the 100-sample gap is dropped; the earlier peak retains precedence.
A5	Decimal Rounding	All floating-point results round to 1 decimal place using half-up arithmetic consistently.
A6	Dual Mode Results	Both `offline` and `online` result lines are stored and outputted; the `mode` key disambiguates them.
A7	Window Grace Period	Online processor waits at most 3.0 s for missing window data before emitting a partial window.
A8	Power Cut Bound	Worst-case power-cut data loss bound is 400 samples/sensor (1 second batch commit interval).
A9	Session Unit	Export and processing sessions are scoped by the mock's 32-bit hexadecimal `session_id`.

Implementation Notes: (1) Mock source: developed and validated against exact protocol test double (`fake_mock.py`). (2) Ground truth verification: validated against analytical ramp formula in `compare_export.py` with 0 discrepancies. (3) AI tools disclosed in `AI_DISCLOSURE.md`.